



◀ COASTAL WATERS

The shallow coastal waters of many cool oceans support lush beds of seaweed that are very efficient producers of food. Some of the most spectacular of these are the kelp forests that grow near the Pacific shores of North America. Here, giant kelp grows in water up to 165 ft (50 m) deep, yet extends its long fronds right up to the surface. The kelp fronds are grazed upon by armies of sea urchins, and these are eaten by sea otters that use stones to smash open the urchins' spiny shells.

► SEABIRD COLONIES

Vast coastal breeding colonies of seabirds rely on the rich waters of continental shelf seas to supply them with food to feed their young. These breeding colonies are particularly numerous near upwelling zones, such as near the coasts of Peru and Chile, and on the fringes of polar oceans where the spring thaw creates a sudden explosion of life in the cold green waters.



▲ RICH FISHERIES

For centuries the richest fisheries were found in the shallow continental shelf seas, which were also conveniently close to land. Intense fishing over the past century has virtually destroyed many populations of the most popular food fish, such as cod and herring. But these fertile waters are still capable of supporting huge fish shoals, and conservation and fishing restrictions may allow the fish stocks to recover.

